Musical ramblings

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1 The basic rules of harmonies

1.1 A bit of physics

[Write this later, not bothered right now]

A sound wave has frequency f, overtones are $2f, 3f, \ldots$.

Equivalently, we are looking at wavelengths λ, and then overtones have wavelengths $\frac{1}{n}\lambda$. Two tones are harmonious if they share a similar overtone, i.e. $\lambda_1 = \frac{n}{m}\lambda_2$.

The fascinating thing seems to be that physics seem to "like" smaller fractions, and we humans can even seemingly hear how "'complex" this fraction is.

1.2 Feeling and Maths

Thus, all harmonious tones (relative to a base tone), are in bijection with $\mathbb{Q}^+$. Further, applying the "harmonies" in succession results in a tone which is still harmonious to our base tone by $(n_1/m_2)(n_2/m_2)\lambda_1 = \lambda_2$. Thus, the group $\mathbb{Q}^+$ (under multiplication) acts freely on tones (i.e. wavelengths or frequencies), and all the orbits are essentially all in "harmony". Thus, for a fixed base tone λ, we analyze its orbit.

2 Western music

Western music is really a human invention. It is a system we use to try and construct "music", which can (somewhat ugly) be reduced to a combination of harmonies and rythm I guess. We now go on to show how one could derive the western musical system from the stuff above, and in doing so, we will see many core concepts of music simply actually being very mathematical.

The first rule of western music is the following:

Rule 2.0.1

Consider $\langle 2 \rangle = \{\dots, 1/4, 1/2, 1, 2, 4, \dots\} \subset \mathbb{Q}^+$. We identify $\lambda_1 \sim \lambda_2$ if they are in the same orbit of $\langle 2 \rangle$.

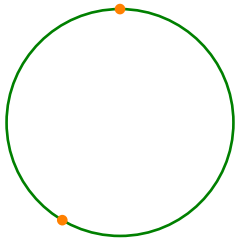
The intervals between tones given by repeated application of 2 (or really 1/2 if talking about wavelengths) is called an octave.

Thus, we are essentially now looking at harmonies as $\mathbb{Q}^+/\langle 2 \rangle$. A nice trick to do here, is now to take the logarithm with base 2

$$\log_2 : \mathbb{Q}^+/\langle 2 \rangle \hookrightarrow \mathbb{R}/\mathbb{Z}$$

We will probably want to specify a finite number of tones that we use. But which ones?

Lets use the unit circle as representatives of the latter set. Given that $(3/1)\lambda \sim (3/2)\lambda$ is the second "nicest" harmony, we will probably want those tones to be in our musical system. Drawn on the unit circle, that tone falls here



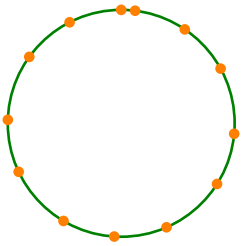
Next, we introduce a rather arbitrary rule, which again gouverns western music.

Rule 2.0.2

Our notes should be evenly spaced (in $\mathbb{R}/\mathbb{Z}$).

One way to justify this rule, is saying that if you want a certain harmony in your note-system (say (3/2) - a perfect fifth), then since the system is not supposed to be made with any fixed base tone, you better also include everything generated by (3/2). But this is more of a "moral" justification - non-western music does not necessarily have this feature!

We quickly see that we will already run into problems though. Repeated application (12 times) of (3/2) leads to the following picture:



Oh god, the horror! Of course, no power of (3/2) can ever be an element of $\langle 2 \rangle$. But we see that after applying (3/2) 12 times, we are almost back where we started. Therefore, if we want

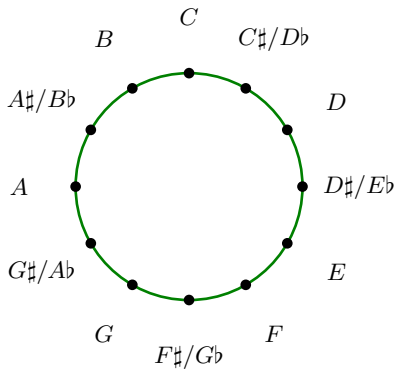
- Our tones should the interval (3/2).

- Rule 2 should be satisfied.

there is just no hope; however, we can somewhat roughly approximate it, by taking our tonal interval to be 1/12-th of the circle (or equivalently, $2^{1/12}$. Its funny how well this works - $2^{1/12}$ is not a fraction, hence $2^{1/12}\lambda_1 = \lambda_2$ don't harmonize at all - yet it is similar enough to $9/8\lambda_1$ to sound okay to us.

Somewhat arbitrarily (for now), we give the tones the following names.

Thus, we get the tones on which all of western music is based, here written with their (for now somewhat arbitrary names).



2.1 Modes

It turns out that playing random combinations of these 12 notes sound quite dissonant, thus we will want to restrain ourselves. Therefore, we will create the arbitrary concept of a **scale**, which we decide

Rule 2.1.1

A scale consists of 7 notes.

(of course, most people would say that a scale consists of 8 notes for some reason, but the first and last are always the root).

Now which seven notes should we pick? Well, our whole tonal system was is based around major-fourths and fifths, so probably we should use those. Thus starting from the root, we have exactly 7 choices: we can apply x fifths from the root, and $7 - x$ fourths from the root. All of these choices appear in western music, and go under the name of "modes", with each scale having a separate name.



Figure 1: The Lydian mode



Figure 2: The Ionian mode - the major scale!



Figure 3: The Mixolydian mode



Figure 4: The Dorian mode



Figure 5: The Aeolian mode - the minor scale!



Figure 6: The Phrygian mode

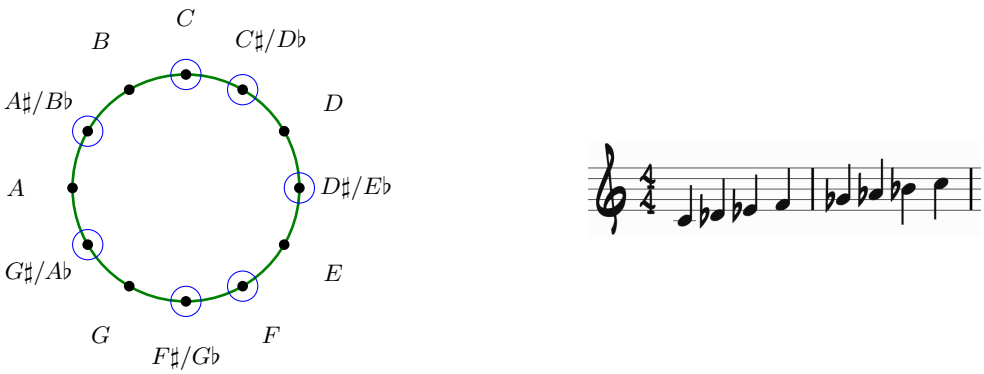


Figure 7: The Locrian mode